

ETMS.Api – Endpoints and Data Structures

Scope: every HTTP endpoint implemented in the ETMS.Api source tree (D:\Sreejith\ETMS_API, working tree of 2026-09-28) and every request / response structure they exchange. Nothing else. **Production base URL:** <https://api.vishwadharma.org> – all full routes below are relative to it. **Sources:** `src/ETMS.Api/Controllers/*.cs`, `src/ETMS.Api.Contracts/**/*.cs`, `src/ETMS.Api/Infrastructure/ApiProblems.cs`, `RateLimiting.cs`, `ScannerDeviceAuthentication.cs`, `src/ETMS.Api.Application/Options/ApiOptions.cs`, `docs/openapi/etms-api-v1.json` (29 paths; all covered).

Source versus deployed API. The binaries in production were published on 2026-09-27 14:51 from this working tree and no source file has changed since, so **the deployed API is this source**. Production configuration differs from the code defaults in one way that matters to the scanner app: `Scanner:RequireDeviceCredential` is `false` in production (pool variable) while the code default is `true`. Everything below describes production behaviour and says where the default would differ.

Production status legend

Label	Meaning
production: verified	called on https://api.vishwadharma.org on 2026-09-27/28 and answered as the source predicts.
production status unverified	present in source and OpenAPI but not confirmable without a JWT (an anonymous call to a JWT route and to a non-existent path give the identical 401 Unauthorized). No difference is expected.

1. Conventions

Item	Rule
Versioning	URL segment. Every route starts with <code>/api/v1/</code> (except <code>/health/</code>).
Transport	HTTPS only. Allowed verbs: GET, POST, PATCH, OPTIONS, HEAD. Plain <code>http://</code> is not served on the public host.
Request body	Content-Type: <code>application/json</code> (UTF-8). Maximum body size 256 KB. A POST without parameters must still carry a Content-Length (send <code>{}</code>), otherwise IIS answers 411.
Response body	<code>application/json</code> on success; <code>application/problem+json</code> on every error.
Property names	camelCase. Enum values are strings , spelled exactly as listed. Success responses emit null properties ; problem responses omit them.
Timestamps	Every <code>DateTime</code> is UTC ISO 8601 with milliseconds and Z: <code>2026-10-03T16:14:05.000Z</code> . Client-sent timestamps may omit milliseconds; a value without Z is treated as UTC. <code>firstEntryLocal</code> is the only local-time value (event time zone Africa/Kampala, format <code>h:mm tt</code> , e.g. <code>7:14 PM</code>).
Identifiers	Tickets: <code>ticketId</code> (GUID string). Users: GUID string. Scanner devices: integer <code>deviceId</code> . Dinner passes: integer <code>passId</code> and <code>passNumber</code> . Gates (officer API only): gate code.
X-Correlation-ID	Optional request header (letters, digits, <code>-</code> ; ≤ 64). Echoed as response header <code>X-Correlation-ID</code> and as <code>correlationId</code> in every body. Generated by the server when absent.
X-Device-Label	Optional request header on scanner routes: free label of the phone (≤ 80 chars after sanitising to letters, digits, space, <code>.</code> , <code>-</code> , <code>_</code> , <code>:</code>). Recorded, with the client IP, as the device name in audit rows and in <code>GET /api/v1/scanner/device</code> .
Rate limiting	Per policy (see each endpoint); 429 <code>RateLimited</code> with <code>Retry-After</code> . Scanner routes without a device token are limited per client IP . Global ceiling 600 requests / minute / IP.

1.1 Access to the API

Group	What the caller sends	Endpoints
Scanner app – no credential (production)	nothing beyond the JSON body; optional X-Device-Label	all /api/v1/scanner/* and /api/v1/corporate-dinner-passes* routes (sections 2.2, 3.4, 4, 5.1, 6, 7, 8.1). The server treats the caller as an unregistered device: deviceId 0, all three scopes (Ticket, GenZ, DinnerPass), no expiry. Ticket admissions are attributed to the virtual gate MOBILE; the app never selects or sees a gate.
Scanner app – optional device token	X-Device-Token: etmsd1.<43 url-safe chars>	the same routes. Only relevant if device registration is switched back on (Scanner:RequireDeviceCredential = true, the code default): then the header is mandatory and comes from POST /api/v1/scanner/enrol. In production a header that is sent is still validated (unknown/revoked token → 401 InvalidDeviceToken); an app that does not send it is never affected.
Officer JWT	Authorization: Bearer <accessToken> (HS256, 15 min, from POST /api/v1/auth/login)	sections 2.1, 2.3, 3.1–3.3, 3.5–3.7, 5.2, 5.3, 8.2 – the earlier officer app and administration. Roles Administrator (ETMS SuperAdmin/Admin) and GateScanner; policies Read / Scan = both roles, Administrator = Administrator only.
none	–	/health/*, POST /api/v1/auth/login, POST /api/v1/auth/refresh, POST /api/v1/scanner/enrol

1.2 Error response structure (all non-2xx)

```
{
  "type": "https://tickets.example.com/problems/ValidationFailed",
  "title": "One or more validation errors occurred.",
  "status": 400,
  "instance": "/api/v1/scanner/tickets/scan",
  "errors": { "qrPayload": ["The QrPayload field is required."] },
  "code": "ValidationFailed",
  "correlationId": "docs-example-0001"
}
```

Field	JSON type	Nullable	Meaning
type	string	no	https://tickets.example.com/problems/<code> – identifier only; the host is a placeholder and does not resolve
title	string	no	human-readable summary
status	integer	no	HTTP status
detail	string	omitted when null	extra detail; Development builds only
instance	string	no	request path
code	string	no	stable machine-readable code (9.1)
correlationId	string	no	
errors	object of string → string[]	only on 400 ValidationFailed	field name (camelCase) → messages

503 responses carry Retry-After (3 from scan endpoints, 5 from the generic handler). 401 responses carry WWW-Authenticate (ScannerDevice realm="ETMS.Api" on scanner routes, Bearer realm="ETMS.Api" on JWT routes).

1.3 Health (no authentication) – production: verified

Method	Route	Response
GET	https://api.vishwadharma.org/health/live	200, text/plain Healthy
GET	https://api.vishwadharma.org/health/ready	200 Healthy or 503 Unhealthy (text/plain)

2. Authentication and device access

2.1 Officer authentication (JWT) – not used by the scanner app

2.1.1 POST `https://api.vishwadharm.org/api/v1/auth/login` – **production: verified**

Sign in with ETMS web credentials; receive a 15-minute access token and a rotating 7-day refresh token bound to `deviceId`.
Auth: none. Rate limit 10 / min / IP; lockout after 5 failures for 15 minutes.

Request `LoginRequest`:

Field	Type	Required	Limits
<code>username</code>	string	yes	≤ 256 (the ETMS e-mail address)
<code>password</code>	string	yes	≤ 256
<code>deviceId</code>	string	yes	4–100 chars, <code>^[A-Za-z0-9._:-]{4,100}\$</code> (installation id of the officer app; unrelated to scanner devices)

Responses: **200** `AuthResponse` (9.3). **400** `ValidationFailed`. **401** `InvalidCredentials`, `AccountLocked`, `AccountInactive`. **403** `PasswordChangeRequired`, `RoleNotPermitted`. **429**.

2.1.2 POST `https://api.vishwadharm.org/api/v1/auth/refresh` – **production: verified**

Exchange a refresh token for a new token pair; the presented token is consumed. Auth: none. Rate limit 30 / min / IP. Request `RefreshRequest`: `refreshToken` string ≤ 512 required; `deviceId` string ≤ 100 required (must equal the login value).

Responses: **200** `AuthResponse`. **400**. **401** `InvalidRefreshToken`, `RefreshTokenExpired`, `RefreshTokenReused`, `DeviceMismatch`, `AccountInactive`. **429**.

2.1.3 POST `https://api.vishwadharm.org/api/v1/auth/logout` – **production status unverified**

Revoke the caller's refresh token(s) and deny the current access token. Auth: JWT. Body optional `LogoutRequest`: `refreshToken` string ≤ 512 (optional), `allDevices` boolean (default false). Responses: **204**. **401**.

2.1.4 GET `https://api.vishwadharm.org/api/v1/auth/me` – **production status unverified**

Profile of the authenticated officer, the device bound to the token and the active event. Auth: JWT. Responses: **200** `MeResponse` (9.3). **401**.

2.1.5 POST `https://api.vishwadharm.org/api/v1/auth/sessions/revoke` – **production status unverified**

Administrator: terminate every session of a user. Auth: JWT Administrator. Rate limit 30 / min. Request `RevokeSessionsRequest`: `userId` uuid required; `reason` string ≤ 200 optional. Responses: **200** `RevokeSessionsResponse` (9.3). **400**. **401**. **403** `Forbidden`. **404** `NotFound`.

2.2 Scanner app access (no login, no registration)

In production the scanner app needs **no call at all before scanning**. The three routes below exist for device identity; only 2.2.2 is useful to the app in open scanning mode. 2.2.1 and 2.2.3 belong to the optional device-registration mechanism, which is implemented but switched off.

2.2.1 POST `https://api.vishwadharm.org/api/v1/scanner/enrol` – **production: verified (answers 401 for an unknown code); not needed in production**

Exchange an administrator-issued enrolment code for a device token. Only meaningful when device registration is switched back on. Auth: none. Rate limit 10 / min / IP.

Request `ScannerEnrolRequest`:

Field	Type	Required	Limits
<code>enrolmentCode</code>	string	yes	8–60; the 12-character code (dashes, spaces, case ignored) or the QR text <code>ETMSDEV.<code></code>

Field	Type	Required	Limits
deviceLabel	string	no	≤ 100
platform	string	no	≤ 50
appVersion	string	no	≤ 30

Responses: **200** ScannerEnrolResponse (9.6). **400. 401** InvalidEnrolmentCode. **429**.

2.2.2 GET https://api.vishwadharma.org/api/v1/scanner/device – production: verified (200 without any credential)

The caller's device identity, the active event, its time zone, the scanner settings and the server time. Auth: none (optional X-Device-Label, optional X-Device-Token). Rate limit 60 / min / IP. Responses: **200** ScannerDeviceInfo (9.6). **401** only when an invalid X-Device-Token was sent.

```
GET /api/v1/scanner/device HTTP/1.1
Host: api.vishwadharma.org
Accept: application/json
X-Device-Label: Pixel 8 - main gate
```

```
{ "deviceId": 0, "deviceName": "Pixel 8 - main gate @203.0.113.10", "scopes": ["Ticket", "GenZ", "DinnerPass"], "eventName": "INDIA DAY 2026",
  "eventStartsAtUtc": "2026-10-03T14:45:00.000Z", "eventTimeZoneId": "Africa/Kampala",
  "settings": { "genZAccessStartTime": "22:30", "genZSingleEntry": false, "genZRequiresTicketEntry": false, "enforceGenZAccessStartTime": true, "syncBatchMax": 200, "downloadPageMax": 1000 },
  "tokenExpiresAtUtc": null, "serverTimeUtc": "2026-10-03T09:00:00.000Z", "correlationId": "3fbe35c26f7a43e2a340e6662eff4fe9" }
```

2.2.3 POST https://api.vishwadharma.org/api/v1/scanner/device/rotate – production: verified (404 without a credential)

Rotate a registered device's token (previous token valid 10 more minutes). Auth: registered X-Device-Token. Body {}. Rate limit 30 / min. Responses: **200** ScannerRotateResponse (9.6). **401** InvalidDeviceToken. **404** NotFound when the caller has no registered credential – the production case.

2.3 Scanner device administration (officer JWT) – production status unverified

Provisioning of registered devices; not used while open scanning mode is on. Auth: JWT **Administrator**; rate limit 30 / min; responses wrapped in ItemResponse<T> (9.2).

#	Method	Route	Purpose	Request	Responses
2. 3. 1	POST	https://api.vishwadharma.org/api/v1/scanner-devices	create a device, receive its single-use enrolment code	CreateScannerDeviceRequest (9.6)	200 ItemResponse<ScannerDeviceEnrolmentResponse>; 400 (errors.gateCode, errors.expiresAtUtc); 401; 403
2. 3. 2	GET	https://api.vishwadharma.org/api/v1/scanner-devices	devices of the active event	–	200 ItemResponse<ScannerDeviceAdminDto[]>; 401; 403
2. 3. 3	POST	https://api.vishwadharma.org/api/v1/scanner-devices/{deviceId}/revoke	revoke a device	path deviceId integer; body RevokeScannerDeviceRequest : reason string 3–200 required	200 ItemResponse<ScannerDeviceAdminDto>; 400; 401; 403; 404 NotFound
2. 3. 4	POST	https://api.vishwadharma.org/api/v1/scanner-devices/{deviceId}/re-enrol	new enrolment code for an existing device	path deviceId; optional body ReEnrolScannerDeviceRequest: enrolmentMinutes integer 5–1440	200 ItemResponse<ScannerDeviceEnrolmentResponse>; 401; 403; 404; 409 InvalidStatusTransition

3. Ticket data

3.1 GET <https://api.vishwadharma.org/api/v1/tickets> – officer JWT (Read) – production status unverified

Paged, filterable list of the active event's tickets; purchaser phone / e-mail masked for gate scanners. Rate limit 240 / min.

Query (`TicketListQuery`, all optional, AND-combined):

Parameter	Type	Default	Allowed values / limits
page	integer	1	≥ 1
pageSize	integer	50	1–100 (outside → 400)
search	string $\leq$ 100	–	ticket number, attendee, purchaser name, e-mail or phone
ticketNumber	string $\leq$ 30	–	
customerName, attendeeName	string $\leq$ 200	–	
phone	string $\leq$ 30	–	
email	string $\leq$ 256	–	
categoryId	integer	–	
status	string	–	Pending, Approved, Issued, Used, Cancelled, Replaced, Rejected, Expired
gateId	string $\leq$ 20	–	gate code of the live admission (officer API concept); unknown → 400
paymentStatus	string	–	Paid, PartiallyPaid, Pending, Waived
issuedFrom, issuedTo	timestamp	–	issuedTo after issuedFrom
scannedFrom, scannedTo	timestamp	–	scannedTo after scannedFrom
sortBy	string	issueDate	issueDate, scannedDate, ticketNumber, attendeeName, customerName, status, category
sortDirection	string	desc	asc, desc

Responses: **200** `PagedResponse<TicketSummaryDto>` (9.2, 9.4). **400** `ValidationFailed`. **401**. **403**.

3.2 GET <https://api.vishwadharma.org/api/v1/tickets/{ticketId}> – officer JWT (Read) – production status unverified

Ticket detail by GUID. Path `ticketId` GUID. Responses: **200** `ItemResponse<TicketDetailDto>` (9.4). **401**. **404** `NotFound`.

3.3 GET <https://api.vishwadharma.org/api/v1/tickets/number/{ticketNumber}> – officer JWT (Read) – production status unverified

Ticket detail by printed number (lookup only). Path `ticketNumber` string $\leq$ 30. Responses: **200** `ItemResponse<TicketDetailDto>`. **400**. **401**. **404**.

3.4 GET <https://api.vishwadharma.org/api/v1/scanner/tickets> – scanner app; production: verified (200 without any credential)

Download of the active event's tickets for the scanner app – initial load and incremental refresh. No contact data, no QR payload, no gate. Auth: none. Rate limit 60 / min / IP.

Parameter	Type	Default	Allowed values
cursor	string	none (= from the beginning)	opaque value from a previous <code>nextCursor</code> ; malformed → 400 <code>errors.cursor</code>
limit	integer	500	1–1000; values outside are clamped , not rejected

Responses: **200** `ScannerSyncPage<ScannerTicketRecord>` (9.2, 9.7). **400** `ValidationFailed`. **401** only with an invalid `X-Device-Token`.

```
GET /api/v1/scanner/tickets?limit=1000 HTTP/1.1
Host: api.vishwadharma.org
Accept: application/json
```

```
{ "items": [
  { "ticketId": "0f8fad5b-d9cb-469f-a165-70867728950e", "ticketNumber": "INDIAD-X7K9-P2MQ-4H8N", "attendeeName": "Sample Attendee", "categoryCode": "VVIP", "categoryName": "VVIP",
    "ticketType": "Paid", "tableNumber": "VV-012", "status": "Issued", "eligibleForEntry": true,
    "entry": { "status": "NotEntered", "atUtc": null }, "genZEntitled": true, "genZStatus": "Active", "genZEntry": { "status": "NotEntered", "atUtc": null } },
  { "ticketId": "7c9e6679-7425-40de-944b-e07fc1f90ae7", "ticketNumber": "INDIAD-Q2WE-8RTY-1UIO", "attendeeName": "Another Sample", "categoryCode": "REGULAR", "categoryName": "Regular",
    "ticketType": "Paid", "tableNumber": null, "status": "Used", "eligibleForEntry": false,
    "entry": { "status": "Entered", "atUtc": "2026-10-03T15:02:41.000Z" }, "genZEntitled": false, "genZStatus": "NotIncluded", "genZEntry": { "status": "NotEntered", "atUtc": null } } ],
  "nextCursor": "dJE6MTIzNDU2Nzg5OjQyMQ", "hasMore": true, "serverTimeUtc": "2026-10-03T09:05:00.000Z", "correlationId": "..." }
```

Next page: the same URL with `&cursor=dJE6MTIzNDU2Nzg5OjQyMQ`, until `hasMore` is `false`. Later refresh: the same URL with the cursor stored after the last complete run; only records changed since are returned (status, entry and Gen-Z entry changes, cancellations). Items are upserts keyed by `ticketId`.

3.5 PATCH <https://api.vishwadharma.org/api/v1/tickets/{ticketId}/status> – officer JWT (Administrator) – production status unverified

Administrator status change with optimistic concurrency. Path `ticketId` GUID. Rate limit 30 / min. Request `UpdateTicketStatusRequest`: `status` string ≤ 20 required (Cancelled from Pending/Approved/Issued, Expired from Issued, Issued from Expired; never Used); `reason` string 5–500 required; `rowVersion` string ≤ 64 required. Responses: **200** `ItemResponse<TicketStatusChangeResponse>` (9.4). **400**. **401**. **403**. **404**. **409** `InvalidStatusTransition`, `ConcurrencyConflict`.

3.6 POST <https://api.vishwadharma.org/api/v1/tickets/{ticketId}/entry-reversal> – officer JWT (Administrator) – production status unverified

Reverse an admission (Used → Issued). Path `ticketId` GUID. Request `EntryReversalRequest`: `reason` string 5–500 required. Responses: **200** `ItemResponse<TicketStatusChangeResponse>`. **400**. **401**. **403**. **404**. **409** `NoLiveAdmission`.

3.7 Reference data – officer JWT (Read) – production status unverified

Meth od	Route	Purpose	200 body
GET	https://api.vishwadharma.org/api/v1/ticket-categories	ticket categories of the active event	<code>ItemResponse<TicketCategoryDto[]></code> (9.4)
GET	https://api.vishwadharma.org/api/v1/gates	gates of the active event (officer app only; the scanner app has no gate)	<code>ItemResponse<GateDto[]></code> (9.4)

4. Corporate dinner pass data – scanner app

Both routes return the same page structure. Auth: none (optional `X-Device-Label`). Rate limit 60 / min / IP. Query: `cursor` and `limit` exactly as in 3.4.

#	Method	Route	Status
4.1	GET	https://api.vishwadharma.org/api/v1/corporate-dinner-passes?cursor=&limit=	production: verified (200 without any credential)
4.2	GET	https://api.vishwadharma.org/api/v1/scanner/dinner-passes?cursor=&limit=	production: verified (alias, same code)

Purpose: download of the individual Corporate Dinner passes of the active event (pass id and number, recipient name, batch, status, entry state with the original entry time). The QR token is never included. Responses: **200** ScannerSyncPage<ScannerDinnerPassRecord> (9.7). **400** ValidationFailed (errors.cursor). **401** only with an invalid X-Device-Token.

```
{ "items": [
  { "passId": 1042, "passNumber": "CDP-7K3M2Q9X", "recipientName": "Sample Guest", "batchReference": "CDP-7K3M-2Q9X", "status": "Issued", "entry": { "status": "NotEntered", "atUtc": null } },
  { "passId": 1043, "passNumber": "CDP-9QX2M7KA", "recipientName": "Second Sample", "batchReference": "CDP-7K3M-2Q9X", "status": "Used", "entry": { "status": "Entered", "atUtc": "2026-10-03T16:20:41.000Z" } } ],
  "nextCursor": "djE6MTIzNDU2Nzg5OjA", "hasMore": false, "serverTimeUtc": "2026-10-03T16:30:00.000Z", "correlationId": "..." }
```

5. Ticket scanning

5.1 POST https://api.vishwadharma.org/api/v1/scanner/tickets/scan – scanner app; production: verified (processes without any credential)

Verify an event ticket QR (ETMS....) and record the regular admission (ticket Issued → Used) atomically at server time. Auth: none (optional X-Device-Label). Rate limit 60 per 10 s per IP. Request ScannerScanRequest (9.5). Responses: **200** ScannerScanResponse for every processed outcome; **400** ValidationFailed; **401** only with an invalid X-Device-Token; **409** ScannerScanResponse with result: "DuplicateRequest"; **429**; **503** ScannerScanResponse with result: "TemporarilyUnavailable" and Retry-After: 3. Result values: Valid, AlreadyUsed, Invalid, WrongCredentialType (a dinner pass QR), Cancelled, Replaced, NotIssued, NotPaid, WrongEvent, Expired.

First scan of a ticket:

```
POST /api/v1/scanner/tickets/scan HTTP/1.1
Host: api.vishwadharma.org
Content-Type: application/json
X-Device-Label: Pixel 8 - main gate
```

```
{ "qrPayload": "ETMS.<body>.<signature>", "clientScanId": "9b2d0d3e-5c4a-4f0a-9d7e-2a1b3c4d5e6f", "scannedAtUtc": "2026-10-03T16:14:03Z" }
```

```
{ "scanType": "Ticket", "result": "Valid", "admissionGranted": true, "replayed": false, "message": "Ticket valid. Entry granted.",
  "firstEntryAtUtc": "2026-10-03T16:14:05.000Z", "firstEntryLocal": "7:14 PM", "reEntryAllowed": false, "accessStartTime": null,
  "credential": { "kind": "Ticket", "ticketNumber": "INDIAD-X7K9-P2MQ-4H8N", "passNumber": null, "attendeeName": "Sample Attendee", "category": "VVIP",
    "ticketType": "Paid", "tableNumber": "VV-012", "status": "Used", "genZEntitled": true, "batchReference": null },
  "clientScanId": "9b2d0d3e-5c4a-4f0a-9d7e-2a1b3c4d5e6f", "processedAtUtc": "2026-10-03T16:14:05.000Z", "correlationId": "..." }
```

New scan of the same ticket later (any phone, new clientScanId) – AlreadyUsed with the original entry time:

```
{ "scanType": "Ticket", "result": "AlreadyUsed", "admissionGranted": false, "replayed": false, "message": "Already scanned at 7:14 PM.",
  "firstEntryAtUtc": "2026-10-03T16:14:05.000Z", "firstEntryLocal": "7:14 PM", "reEntryAllowed": false, "accessStartTime": null,
  "credential": { "kind": "Ticket", "ticketNumber": "INDIAD-X7K9-P2MQ-4H8N", "passNumber": null, "attendeeName": "Sample Attendee", "category": "VVIP",
    "ticketType": "Paid", "tableNumber": "VV-012", "status": "Used", "genZEntitled": true, "batchReference": null },
  "clientScanId": "4c1d2e3f-6a7b-4c8d-9e0f-1a2b3c4d5e70", "processedAtUtc": "2026-10-03T16:31:12.000Z", "correlationId": "..." }
```

Retry of the first request (same `clientScanId`, same QR) – the stored result is replayed, nothing is recorded again:

```
{ "scanType": "Ticket", "result": "Valid", "admissionGranted": true, "replayed": true, "message": "Ticket valid. Entry granted.",
  "firstEntryAtUtc": "2026-10-03T16:14:05.000Z", "firstEntryLocal": "7:14 PM", "reEntryAllowed": false, "accessStartTime": null,
  "credential": { "kind": "Ticket", "ticketNumber": "INDIAD-X7K9-P2MQ-4H8N", "passNumber": null, "attendeeName": "Sample Attendee", "category": "VVIP",
    "ticketType": "Paid", "tableNumber": "VV-012", "status": "Used", "genZEntitled": true, "batchReference": null },
  "clientScanId": "9b2d0d3e-5c4a-4f0a-9d7e-2a1b3c4d5e6f", "processedAtUtc": "2026-10-03T16:14:05.000Z", "correlationId": "..." }
```

Same `clientScanId` with a *different* QR or scan type → **409**, body `"result": "DuplicateRequest", "credential": null`.

5.2 POST <https://api.vishwadharm.org/api/v1/ticket-scans> – officer JWT (Scan) – production status **unverified; not for the scanner app**

Officer app: validate a ticket QR at a named gate. Rate limit 30 per 10 s per user. Request `TicketScanRequest` (9.5) – requires `gateId` and `deviceId`, which is why the scanner app must not use it. Responses: **200** `TicketScanResponse` (9.5) for `Valid`, `AlreadyUsed`, `Invalid`, `Cancelled`, `Replaced`, `NotPaid`, `NotIssued`, `WrongEvent`, `Expired`, `UnauthorizedGate`. **400. 401. 403. 409** body with `result: "DuplicateRequest"`. **422** `GateNotFound`, `DeviceMismatch`. **429. 503** body with `result: "TemporarilyUnavailable"`.

5.3 GET <https://api.vishwadharm.org/api/v1/scans/recent> – officer JWT (Read) – production status **unverified**

Recent officer scans, newest first (gate scanners: own scans; administrators: all). Rate limit 240 / min. Query `RecentScansQuery`: `page ≥ 1` (1); `pageSize 1–100` (50); `gateId string ≤ 20` (existing gate code); `result string ≤ 30` (`ScanResultCode` name); `admissionsOnly boolean`; `from, to timestamps` (to after from); `userId uuid` (administrators only). Responses: **200** `PagedResponse<RecentScanDto>` (9.5). **400. 401. 403.**

6. Gen Z scanning – scanner app

6.1 POST <https://api.vishwadharm.org/api/v1/scanner/genz/scan> – production: verified (processes without any credential)

Verify the same event ticket QR and record the Gen-Z Dancing Area entry, independently of the regular entry. Auth: none. Rate limit 60 per 10 s per IP. Request `ScannerScanRequest` (9.5). Responses as 5.1. Additional result values: `NotEligible` (no paid, active Gen-Z entitlement), `BeforeStartTime` (before `settings.genZAccessStartTime`, event local time; `accessStartTime` returned), `EntryRequired` (only if the server were configured to require main-gate entry first; it is not).

First Gen-Z scan (`Valid`):

```
{ "scanType": "GenZ", "result": "Valid", "admissionGranted": true, "replayed": false, "message": "Gen-Z access granted.",
  "firstEntryAtUtc": "2026-10-03T20:02:03.000Z", "firstEntryLocal": "11:02 PM", "reEntryAllowed": false, "accessStartTime": "22:30",
  "credential": { "kind": "Ticket", "ticketNumber": "INDIAD-X7K9-P2MQ-4H8N", "passNumber": null, "attendeeName": "Sample Attendee", "category": "VVIP",
    "ticketType": "Paid", "tableNumber": "VV-012", "status": "Used", "genZEntitled": true, "batchReference": null },
  "clientScanId": "a1b2c3d4-e5f6-4a7b-8c9d-0e1f2a3b4c5d", "processedAtUtc": "2026-10-03T20:02:03.000Z", "correlationId": "..." }
```

New scan of the same ticket at Gen-Z (new `clientScanId`) – AlreadyUsed with the original Gen-Z entry time; `reEntryAllowed` is true while the organisers allow re-entry (`settings.genZSingleEntry` = false):

```
{ "scanType": "GenZ", "result": "AlreadyUsed", "admissionGranted": false, "replayed": false, "message": "Already scanned at 11:02 PM - re-entry permitted.",
  "firstEntryAtUtc": "2026-10-03T20:02:03.000Z", "firstEntryLocal": "11:02 PM", "reEntryAllowed": true, "accessStartTime": "22:30",
  "credential": { "kind": "Ticket", "ticketNumber": "INDIAD-X7K9-P2MQ-4H8N", "passNumber": null, "attendeeName": "Sample Attendee", "category": "VVIP",
    "ticketType": "Paid", "tableNumber": "VV-012", "status": "Used", "genZEntitled": true, "batchReference": null },
  "clientScanId": "b2c3d4e5-f6a7-4b8c-9d0e-1f2a3b4c5d6e", "processedAtUtc": "2026-10-03T20:40:10.000Z", "correlationId": "..." }
```

Retry with the original `clientScanId`: the Valid response above with `"replayed": true`. Before opening: `"result": "BeforeStartTime", "message": "The Gen-Z area opens at 10:30 PM.", "accessStartTime": "22:30", "credential": { ... "status": "Issued" ... }`. No entitlement: `"result": "NotEligible", "message": "This ticket does not include Gen-Z access."`.

7. Corporate dinner pass scanning – scanner app

Both routes run the same code. Auth: none. Rate limit 60 per 10 s per IP. Request `ScannerScanRequest` with `qrPayload` = `CDP1.<token>`.

#	Method	Route	Status
7.1	POST	<code>https://api.vishwadharm.org/api/v1/corporate-dinner-passes/scans</code>	production: verified (processes without any credential)
7.2	POST	<code>https://api.vishwadharm.org/api/v1/scanner/dinner-passes/scan</code>	production: verified (alias)

Purpose: verify a Corporate Dinner pass QR and record the first dinner entry atomically (pass Issued → Used, server time). Responses as 5.1. Result values: Valid, AlreadyUsed, Invalid (unknown, tampered, or a `CDPSAMPLE`. sample pass), WrongCredentialType (a ticket QR), Cancelled, WrongEvent; plus 409 DuplicateRequest, 503 TemporarilyUnavailable. First scan (Valid):

```
{ "scanType": "DinnerPass", "result": "Valid", "admissionGranted": true, "replayed": false, "message": "Welcome - please enjoy the dinner.",
  "firstEntryAtUtc": "2026-10-03T16:20:41.000Z", "firstEntryLocal": "7:20 PM", "reEntryAllowed": false, "accessStartTime": null,
  "credential": { "kind": "DinnerPass", "ticketNumber": null, "passNumber": "CDP-7K3M2Q9X", "attendeeName": "Sample Guest", "category": null, "ticketType": null,
    "tableNumber": null, "status": "Used", "genZEntitled": null, "batchReference": "CDP-7K3M-2Q9X" },
  "clientScanId": "c3d4e5f6-a7b8-4c9d-0e1f-2a3b4c5d6e7f", "processedAtUtc": "2026-10-03T16:20:41.000Z", "correlationId": "..." }
```

New scan of the same pass (new `clientScanId`, any phone) – AlreadyUsed with the original entry time (there is no re-entry for dinner passes):

```
{ "scanType": "DinnerPass", "result": "AlreadyUsed", "admissionGranted": false, "replayed": false, "message": "Already scanned at 7:20 PM.",
  "firstEntryAtUtc": "2026-10-03T16:20:41.000Z", "firstEntryLocal": "7:20 PM", "reEntryAllowed": false, "accessStartTime": null,
  "credential": { "kind": "DinnerPass", "ticketNumber": null, "passNumber": "CDP-7K3M2Q9X", "attendeeName": "Sample Guest", "category": null, "ticketType": null,
    "tableNumber": null, "status": "Used", "genZEntitled": null, "batchReference": "CDP-7K3M-2Q9X" },
  "clientScanId": "d4e5f6a7-b8c9-4d0e-1f2a-3b4c5d6e7f80", "processedAtUtc": "2026-10-03T17:05:00.000Z", "correlationId": "..." }
```

Retry with the original `clientScanId`: the Valid response with `"replayed": true`. Ticket QR here: `"result": "WrongCredentialType", "message": "This is an INDIA DAY 2026 admission ticket, not a Corporate Dinner pass. Use 'Scan Ticket Entry'." , "credential": null`. Cancelled pass: `"result": "Cancelled", "message": "This pass was cancelled by the organisers and is not valid for entry."`.

8. Synchronisation

8.1 POST <https://api.vishwadharma.org/api/v1/scanner/sync> – scanner app; production: verified (route reachable without credential)

Upload scans captured offline (any mix of the three types); each item receives its own final result. Auth: none. Rate limit 20 / min / IP. Request `ScannerSyncRequest` (9.5): items 1–200 (more → 400 `errors.items`). Responses: **200** `ScannerSyncResponse` (9.5). **400** `ValidationFailed`. **429**. Items are processed in order with exactly the logic of the live endpoints, so a re-sent batch replays stored results (`replayed: true`).

```
POST /api/v1/scanner/sync HTTP/1.1
Host: api.vishwadharma.org
Content-Type: application/json
```

```
{ "items": [
  { "scanType": "Ticket", "qrPayload": "ETMS.<body>.<signature>", "clientScanId": "e5f6a7b8-c9d0-4e1f-2a3b-4c5d6e7f8091", "scannedAtUtc": "2026-10-03T16:10:00Z", "offlineDecision": "Admitted" },
  { "scanType": "DinnerPass", "qrPayload": "CDP1.<encrypted pass token>", "clientScanId": "f6a7b8c9-d0e1-4f2a-3b4c-5d6e7f8091a2", "scannedAtUtc": "2026-10-03T16:11:30Z", "offlineDecision": "Admitted" }
] }
```

```
{ "results": [
  { "clientScanId": "e5f6a7b8-c9d0-4e1f-2a3b-4c5d6e7f8091", "scanType": "Ticket", "accepted": true, "conflict": true,
    "response": { "scanType": "Ticket", "result": "AlreadyUsed", "admissionGranted": false, "replayed": false, "message": "Already scanned at 7:14 PM.",
      "firstEntryAtUtc": "2026-10-03T16:14:05.000Z", "firstEntryLocal": "7:14 PM", "reEntryAllowed": false, "accessStartTime": null,
      "credential": { "kind": "Ticket", "ticketNumber": "INDIAD-X7K9-P2MQ-4H8N", "passNumber": null, "attendeeName": "Sample Attendee", "category": "VVIP", "ticketType": "Paid", "tableNumber": "VV-012", "status": "Used", "genZEntitled": true, "batchReference": null },
      "clientScanId": "e5f6a7b8-c9d0-4e1f-2a3b-4c5d6e7f8091", "processedAtUtc": "2026-10-03T16:30:00.000Z", "correlationId": "..." } },
  { "clientScanId": "f6a7b8c9-d0e1-4f2a-3b4c-5d6e7f8091a2", "scanType": "DinnerPass", "accepted": true, "conflict": false,
    "response": { "scanType": "DinnerPass", "result": "Valid", "admissionGranted": true, "replayed": false, "message": "Welcome - please enjoy the dinner.",
      "firstEntryAtUtc": "2026-10-03T16:30:00.000Z", "firstEntryLocal": "7:30 PM", "reEntryAllowed": false, "accessStartTime": null,
      "credential": { "kind": "DinnerPass", "ticketNumber": null, "passNumber": "CDP-7K3M2Q9X", "attendeeName": "Sample Guest", "category": null, "ticketType": null, "tableNumber": null, "status": "Used", "genZEntitled": null, "batchReference": "CDP-7K3M-2Q9X" },
      "clientScanId": "f6a7b8c9-d0e1-4f2a-3b4c-5d6e7f8091a2", "processedAtUtc": "2026-10-03T16:30:00.000Z", "correlationId": "..." } } ],
  "accepted": 2, "conflicts": 1, "serverTimeUtc": "2026-10-03T16:30:00.000Z", "correlationId": "..." }
```

The entry time of a sync `Valid` is the server time the batch was processed, never `scannedAtUtc`.

8.2 POST https://api.vishwadharm.org/api/v1/ticket-scans/sync – officer JWT (Scan) – disabled by configuration; production status unverified

Officer app batch reconciliation. Request `OfflineSyncRequest`: `scans` array of `TicketScanRequest`, 1–500 (server limit 200). Responses: **501** `OfflineSyncDisabled` in production (`Scanning:OfflineSyncEnabled` = `false`); when enabled 200 `OfflineSyncResponse` (9.5); 400; 401; 403.

9. Data structures

Types: `string`, `integer`, `number`, `boolean`, `timestamp` (UTC ISO 8601, section 1), `uuid`, `object`, `array`. "Nullable" = may be JSON `null`.

9.1 Error codes (code)

Code	HT TP	Where
<code>ValidationFailed</code>	400	any body/query that fails validation; unreadable body
<code>RequestCancelled</code>	400	client aborted
<code>InvalidCredentials</code> , <code>AccountLocked</code> , <code>AccountInactive</code>	401	officer login (<code>AccountInactive</code> also refresh)
<code>InvalidRefreshToken</code> , <code>RefreshTokenExpired</code> , <code>RefreshTokenReused</code> , <code>DeviceMismatch</code>	401	officer refresh
<code>Unauthorized</code>	401	JWT missing/invalid on officer routes; also any unknown path
<code>InvalidEnrolmentCode</code>	401	POST <code>/api/v1/scanner/enrol</code>
<code>InvalidDeviceToken</code>	401	scanner routes: only when an X-Device-Token was sent and is unknown/revoked/expired (or, with registration switched on, when it is missing)
<code>PasswordChangeRequired</code> , <code>RoleNotPermitted</code>	403	officer login
<code>Forbidden</code>	403	JWT role not permitted
<code>ScopeNotPermitted</code>	403	a <i>registered</i> device lacking the scope (an unregistered device has all scopes)
<code>NotFound</code>	404	ticket / user / device not found; rotate without a registered credential
<code>ConcurrencyConflict</code> , <code>InvalidStatusTransition</code> , <code>NoLiveAdmission</code>	409	ticket status change / entry reversal / device re-enrol
<code>DuplicateRequest</code>	409	same <code>clientScanId</code> , different payload or type (body is the scan response)
<code>GateNotFound</code> , <code>DeviceMismatch</code>	422	officer POST <code>/api/v1/ticket-scans</code> only
<code>RateLimited</code>	429	any rate-limited route; <code>Retry-After</code>
<code>InternalServerError</code>	500	unexpected failure
<code>OfflineSyncDisabled</code>	501	POST <code>/api/v1/ticket-scans/sync</code>
<code>ServiceUnavailable</code> , <code>ScannerGateNotConfigured</code>	503	database failure; virtual gate <code>MOBILE</code> missing for the active event

9.2 Envelopes and pagination

`ScannerSyncPage<T>` (cursor pagination – scanner downloads 3.4 and 4)

Field	Type	Nullabl e	Meaning
<code>items</code>	array of T	no	upserts keyed by <code>ticketId</code> / <code>passId</code>

Field	Type	Nullable	Meaning
nextCursor	string	no	opaque; pass as ?cursor= next time. Internally base64url("v1:<rowVersion>:<genZLogId>") – do not parse
hasMore	boolean	no	true ⇒ call again immediately with nextCursor
serverTimeUtc	timestamp	no	
correlationId	string	no	

PagedResponse<T> (page-number pagination – officer GET /api/v1/tickets, GET /api/v1/scans/recent): data array; pagination (page, pageSize, totalRecords, totalPages, hasNextPage, hasPreviousPage – all integers/booleans, not nullable); correlationId. **ItemResponse<T>** – { "data": T, "correlationId": string }.

9.3 Officer authentication structures (JWT routes only)

AuthResponse – accessToken string; refreshToken string; tokenType Bearer; expiresIn integer (900); refreshExpiresIn integer (604800); accessTokenExpiresAtUtc timestamp; user AuthUserDto. **AuthUserDto** – id uuid; name string; username string; role Administrator | GateScanner; defaultGate GateRefDto nullable. **GateRefDto** – id string (gate code), name string. **MeResponse** – user AuthUserDto; roles string[]; deviceId string; accessTokenExpiresAtUtc timestamp; activeEventName string; activeEventStartsAtUtc timestamp. **RevokeSessionsResponse** – userId uuid; refreshTokensRevoked integer; revokedAtUtc timestamp.

9.4 Ticket structures (officer JWT routes)

TicketSummaryDto – ticketId uuid; ticketNumber string; status string (Pending, Approved, Issued, Used, Cancelled, Replaced, Rejected, Expired); ticketType string; category { id integer, code string, name string }; attendee { name string }; customer { name string, phone string nullable (masked for gate scanners), email string nullable (masked), isMasked boolean }; payment { status Paid|PartiallyPaid|Pending|Waived, amount number, currency string, method string nullable }; issueDateUtc timestamp nullable; scannedDateUtc timestamp nullable; gate GateRefDto nullable; tableNumber string nullable. **TicketDetailDto** = summary + bookingReference string; bookingQuantity integer; createdAtUtc timestamp; cancelledAtUtc timestamp nullable; cancelReason string nullable (administrators); replacedByTicketNumber, replacesTicketNumber string nullable; rowVersion string nullable (administrators); history TicketStatusHistoryDto[] nullable (administrators); scans TicketScanHistoryDto[]; genZAccess boolean; genZStatus string (NotSelected, PendingPayment, Active, Cancelled, Refunded, Used); genZAccessStartTime string HH:mm nullable. **TicketStatusHistoryDto** – fromStatus string nullable; toStatus string; changedAtUtc timestamp; reason string nullable. **TicketScanHistoryDto** – scannedAtUtc timestamp; result string; isAdmission boolean; isReversed boolean; gate GateRefDto; officerName string. **TicketStatusChangeResponse** – ticketId uuid; ticketNumber string; previousStatus, newStatus string; changedAtUtc timestamp; changedBy string; reason string; rowVersion string nullable. **TicketCategoryDto** – id integer; code, name string; price number; currency string; ticketType string; isActive boolean; displayColour string; sortOrder integer. **GateDto** – id string; name string; description string nullable; isActive boolean; opensAtUtc, closesAtUtc timestamp nullable; isOpenNow boolean.

9.5 Scan structures

Enum ScanType (scanner app): Ticket, GenZ, DinnerPass. **Enum ScannerResult** (scanner app result): Valid, AlreadyUsed, Invalid, WrongCredentialType, NotEligible, Cancelled, Replaced, NotIssued, NotPaid, WrongEvent, Expired, BeforeStartTime, EntryRequired, DuplicateRequest (HTTP 409), TemporarilyUnavailable (HTTP 503). **Enum ScanResultCode** (officer API result): Valid, AlreadyUsed, Invalid, Cancelled, Replaced, NotPaid, NotIssued, WrongEvent, Expired, UnauthorizedGate, DuplicateRequest, TemporarilyUnavailable.

ScannerScanRequest (body of 5.1, 6.1, 7.x; base of each sync item) – no gate, no operator, no device id

Field	Type	Required	Nullable	Limits / values
qrPayload	string	yes	no	6–600; complete decoded QR text (ETMS.... ticket, CDP1.... dinner pass)
clientScanId	uuid	yes	no	one per physical scan; reused only for retries of that scan

Field	Type	Required	Nullable	Limits / values
scannedAtUtc	timestamp	no	yes	device clock, diagnostic only – never the entry time
offlineDecision	string	no	yes	Admitted, Rejected (other values → 400)

ScannerScanResponse (200 / 409 / 503 of the scan routes; response of each sync result)

Field	Type	Nullable	Meaning / values
scanType	ScanType	no	
result	ScannerResult	no	
admissionGranted	boolean	no	true only when this request (or the original it replays) recorded the first admission
replayed	boolean	no	true when the response is the stored outcome of an earlier request with the same clientScanId (retry); false for a new scan, including a new scan of an already used credential
message	string	no	operator text
firstEntryAtUtc	timestamp	yes	server UTC time of the first admission at this checkpoint: set for Valid (now) and AlreadyUsed (the original time – never changed); null otherwise
firstEntryLocal	string	yes	the same instant in event local time, h:mm tt
reEntryAllowed	boolean	no	Gen-Z only: re-entry permitted although AlreadyUsed
accessStartTime	string HH:mm	yes	Gen-Z only
credential	ScannerCredentialDto	yes	null when the QR could not be resolved
clientScanId	uuid	no	echo
processedAtUtc	timestamp	no	
correlationId	string	no	

ScannerCredentialDto – kind string (Ticket, DinnerPass); ticketNumber string nullable; passNumber string nullable; attendeeName string nullable (attendee or recipient); category, ticketType, tableNumber string nullable (tickets); status string nullable (status **after** the scan); genZEntitled boolean nullable (tickets); batchReference string nullable (passes). No gate field.

ScannerSyncRequest – items ScannerSyncItem[] 1–200. **ScannerSyncItem** = ScannerScanRequest + scanType (ScanType, required). **ScannerSyncResponse** – results ScannerSyncItemResult[]; accepted integer; conflicts integer; serverTimeUtc timestamp; correlationId string. **ScannerSyncItemResult** – clientScanId uuid; scanType ScanType; accepted boolean (final outcome stored – false only for TemporarilyUnavailable); conflict boolean (offlineDecision was Admitted but response.result ≠ Valid); response ScannerScanResponse.

Officer API only – TicketScanRequest (POST /api/v1/ticket-scans): qrPayload string 10–512 required; gateId string ≤ 20 ^[A-Za-z0-9_-]{1,20}\$ required; deviceId string ≤ 100 required (must equal the token's); clientScanId uuid required; scannedAtUtc timestamp optional. **TicketScanResponse** – result ScanResultCode; admissionGranted boolean; message string; ticket ScannedTicketDto nullable; replayed boolean; clientScanId uuid; processedAtUtc timestamp; correlationId string. **ScannedTicketDto** – ticketNumber string; attendeeName, category, ticketType string nullable; status string; tableNumber string nullable; entryTimeUtc timestamp nullable (Valid); gate string nullable; previousEntryTimeUtc timestamp nullable (AlreadyUsed, original time); previousGate string nullable; genZAccess boolean; genZStatus string; genZAccessStartTime string nullable. **OfflineSyncRequest** – scans TicketScanRequest[] 1–500.

OfflineSyncResponse – results TicketScanResponse[]; correlationId. **RecentScanDto** – scanId integer; scannedAtUtc

timestamp; result string; admissionGranted boolean; isReversed boolean; ticketId uuid nullable; ticketNumber, attendeeName, category string nullable; gate GateRefDto; officerName string; deviceId string nullable; message string nullable.

9.6 Scanner device structures

ScannerDeviceInfo (200 of 2.2.2)

Field	Type	Nulla ble	Meaning / production value
deviceId	integer	no	0 for an unregistered caller; the <code>ApiScanDevices</code> id for a registered device
deviceName	string	no	<X-Device-Label or "Open scanner"> @<client IP> for an unregistered caller; the administrator-given name for a registered device
scopes	string[]	no	["Ticket", "GenZ", "DinnerPass"] for an unregistered caller
eventName	string	no	
eventStarts AtUtc	timestamp	no	
eventTimeZo neId	string	no	IANA id, Africa/Kampala
settings	ScannerSetti ngsDto	no	
tokenExpire sAtUtc	timestamp	yes	null for an unregistered caller
serverTimeU tc	timestamp	no	
correlation Id	string	no	

ScannerSettingsDto – genZAccessStartTime string HH:mm; genZSingleEntry boolean; genZRequiresTicketEntry boolean; enforceGenZAccessStartTime boolean; syncBatchMax integer; downloadPageMax integer.

Registration structures (implemented; unused while registration is off): **ScannerEnrolResponse** – deviceId integer; deviceName string; deviceToken string (etmsd1.<43 url-safe chars>, shown once); scopes string[]; tokenExpiresAtUtc timestamp nullable; eventName string; serverTimeUtc timestamp. **ScannerRotateResponse** – deviceToken string; rotatedAtUtc timestamp; previousTokenValidUntilUtc timestamp. **CreateScannerDeviceRequest** – name string 2–100 required; scopes string[] optional (subset of Ticket, GenZ, DinnerPass; empty = all); gateCode string ≤ 20 optional (administrator-side attribution only; must exist for the active event); expiresAtUtc timestamp optional (future); enrolmentMinutes integer 5–1440 optional (default 30). **ScannerDeviceEnrolmentResponse** – deviceId integer; name string; scopes string[]; enrolmentCode string (XXXX-XXXX-XXXX, alphabet ABCDEFGHJKLMNPQRSTUVWXYZ23456789); enrolmentQr string (ETMSDEV.<12 chars>); enrolmentExpiresAtUtc timestamp. **ScannerDeviceAdminDto** – deviceId integer; name string; scopes string[]; gateCode string nullable; status string (PendingEnrolment, EnrolmentExpired, Active, Expired, Revoked); createdAtUtc timestamp; createdBy string nullable; enrolledAtUtc timestamp nullable; deviceLabel, platform, appVersion string nullable; tokenHint string nullable; lastSeenAtUtc timestamp nullable; scanCount integer; expiresAtUtc, revokedAtUtc timestamp nullable; revokedReason string nullable. **RevokeScannerDeviceRequest** – reason string 3–200. **ReEnrolScannerDeviceRequest** – enrolmentMinutes integer 5–1440 optional.

9.7 Download records (scanner app)

EntryStateDto – status string (NotEntered, Entered); atUtc timestamp nullable (server time of the first admission; never changes).

ScannerTicketRecord (items of 3.4) – no gate, no contact data

Field	Type	Nullabl e	Meaning / values
ticketId	uuid	no	upsert key; equals the secure id inside the ticket QR body

Field	Type	Nullabl e	Meaning / values
ticketNumber	string	no	
attendeeName	string	no	the only personal data
categoryCode, categoryName	string	no	
ticketType	string	no	
tableNumber	string	yes	
status	string	no	Pending, Approved, Issued, Used, Cancelled, Replaced, Rejected, Expired
eligibleForEntry	boolean	no	Issued and (paid tickets) fully paid – display hint only
entry	EntryStateDt o	no	regular entry
genZEntitled	boolean	no	
genZStatus	string	no	NotIncluded, PendingPayment, Active, Cancelled, Refunded, Used
genZEntry	EntryStateDt o	no	Gen-Z entry, independent of entry

ScannerDinnerPassRecord (items of 4.1 / 4.2)

Field	Type	Nullable	Meaning / values
passId	integer	no	upsert key
passNumber	string	no	unique
recipientName	string	no	
batchReference	string	no	
status	string	no	Issued, Used, Cancelled
entry	EntryStateDto	no	dinner entry

10. Endpoint index

Method	Route (prefix https://api.vishwadharm.org)	Caller sends	Production status
GET	/health/live, /health/ready	nothing	verified
GET	/api/v1/scanner/device	nothing (optional X-Device-Label)	verified, 200
GET	/api/v1/scanner/tickets	nothing	verified, 200
GET	/api/v1/corporate-dinner-passes	nothing	verified, 200
GET	/api/v1/scanner/dinner-passes	nothing	verified, 200
POST	/api/v1/scanner/tickets/scan	JSON body	verified, 200
POST	/api/v1/scanner/genz/scan	JSON body	verified, 200
POST	/api/v1/corporate-dinner-passes/scans	JSON body	verified, 200
POST	/api/v1/scanner/dinner-passes/scan	JSON body	verified (alias)
POST	/api/v1/scanner/sync	JSON body	verified

Method	Route (prefix <code>https://api.vishwadharm.org</code>)	Caller sends	Production status
POST	<code>/api/v1/scanner/device/rotate</code>	registered X-Device-Token	verified (404 without one)
POST	<code>/api/v1/scanner/enrol</code>	enrolment code	verified (not needed)
POST	<code>/api/v1/auth/login</code>	credentials	verified
POST	<code>/api/v1/auth/refresh</code>	refresh token	verified
POST	<code>/api/v1/auth/logout</code>	JWT	unverified
GET	<code>/api/v1/auth/me</code>	JWT	unverified
POST	<code>/api/v1/auth/sessions/revoke</code>	JWT Administrator	unverified
POST / GET	<code>/api/v1/scanner-devices</code>	JWT Administrator	unverified
POST	<code>/api/v1/scanner-devices/{deviceId}/revoke, .../re-enrol</code>	JWT Administrator	unverified
GET	<code>/api/v1/tickets, /api/v1/tickets/{ticketId}, /api/v1/tickets/number/{ticketNumber}</code>	JWT Read	unverified
PATCH	<code>/api/v1/tickets/{ticketId}/status</code>	JWT Administrator	unverified
POST	<code>/api/v1/tickets/{ticketId}/entry-reversal</code>	JWT Administrator	unverified
GET	<code>/api/v1/ticket-categories, /api/v1/gates</code>	JWT Read	unverified
POST	<code>/api/v1/ticket-scans</code>	JWT Scan (+ gateId, deviceId)	unverified
GET	<code>/api/v1/scans/recent</code>	JWT Read	unverified
POST	<code>/api/v1/ticket-scans/sync</code>	JWT Scan	unverified; returns 501 by configuration

11. Requirements versus implementation (2026-09-28)

Requirement for the scanner app	Implementation
No operator login	Met. No scanner route uses a JWT.
No device registration, verification or credential	Met in production by configuration (<code>Scanner:RequireDeviceCredential = false</code>). Two caveats from the code: (1) an X-Device-Token header that a client <i>chooses</i> to send is still validated and rejected with 401 if unknown – an app that never sends the header is unaffected; (2) the code default is <code>true</code> , so a deployment without the production pool variable would require registration again.
No device matching	Met. No <code>deviceId</code> field exists in any scanner request or response (the <code>deviceId</code> in <code>GET /api/v1/scanner/device</code> is the server's identity value, 0 for every unregistered caller).
No gate selection or gate data	Met. No scanner request, response or download record carries a gate. Admissions are attributed server-side to the virtual gate <code>MOBILE</code> ; if that gate is missing the scan answers 503 <code>ScannerGateNotConfigured</code> .
Separate ticket / Gen-Z / dinner scan endpoints	Met: 5.1, 6.1, 7.1 (7.2 alias).
Download and synchronisation endpoints	Met: 3.4, 4.1 (4.2 alias), 8.1.
AlreadyUsed with the original entry time after the first admission	Met for all three scan types (<code>firstEntryAtUtc</code> , <code>firstEntryLocal</code> , never overwritten). A retry with the same <code>clientScanId</code> is a different case: it replays the stored result with <code>replayed: true</code> .

Requirement for the scanner app	Implementation
OpenAPI file	Descriptions updated to open scanning mode. The <code>security</code> entries of the scanner operations still list the <code>DeviceToken</code> scheme because the file is generated from the source (<code>SwaggerSetup.cs</code>); a client generated from it will treat <code>X-Device-Token</code> as required unless that requirement is removed by hand. Regenerating the file from the current source restores the old wording.